EE345 Mod3-Lec4: Kernel Methods

References: no textbook reference

Motivation for Kernels in Regression

Recall typical setup for data fitting/regression (Module 2):

- features $x_i \in \mathbb{R}^d$, output $y_i \in \mathbb{R}$, $i = 1, \ldots, n$
- goal: find predictor function $f(x)$ such that $f(x_i) \approx y_i$, where

$$f(x) = \sum_{i=1}^{p} \theta_i \phi_i(x)$$

  is **linear** in parameters $\theta \in \mathbb{R}^p$, and $\phi_i(x)$ are **feature maps**
- In particular, $\phi(x) = (\phi_1(x), \ldots, \phi_p(x))$ is a vector of feature maps.

Motivation for Kernels in Regression

- The predictor can be *nonlinear in data* via feature maps $\phi_i(x)$ (cf. Module 2)

$$x \mapsto 1, x, x^2, \ldots \quad \text{or} \quad x \mapsto \max\{0, x - c_i\}$$

- We'd like to make adding new nonlinear features a bit more "automatic" $\implies$ **kernel methods**
- Basic structure:
  - Each row in the data matrix $X \in \mathbb{R}^{n \times d}$ is a sample.
  - ϕ **lifts** each row from $\mathbb{R}^d$ to $\mathbb{R}^p$
  - The kernel matrix is $K \in \mathbb{R}^{n \times n}$, comparing every pair of rows (samples)

Polynomial Kernel Example

Data: $n = 3$ samples in $d = 2$ dimensions:

$$X = \begin{bmatrix} — & x_1^\top & — \\ — & x_2^\top & — \\ — & x_3^\top & — \end{bmatrix}$$

Polynomial features up to degree 2 gives $p = 5$ features:

$$\phi(x) = (x_1, x_2, x_1^2, x_1 x_2, x_2^2)$$

so that

$$f(x) = \theta^\top \phi(x) = \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1^2 + \theta_4 x_1 x_2 + \theta_5 x_2^2.$$

Feature Lifting and the Kernel Trick

- The input feature vector is:

$$x = (x_1, x_2, \ldots, x_d) \in \mathbb{R}^d.$$

- A **feature map** lifts x into a higher-dimensional space:

- The model is **linear in feature space**:

- The **kernel trick** avoids explicit feature lifting:

The Kernel Trick

- The feature dimension p can be huge (or infinite).
- Instead of computing $\phi(x)$ explicitly, use a kernel function:

$$k(x, x') = \phi(x)^\top \phi(x').$$

- Learning depends only on **inner products** in feature space.

Representer Theorem

The optimal f lies in the span of training features

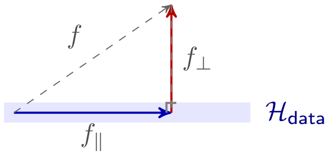
$$f(x) = \sum_{i=1}^{n} \alpha_i \, k(x, x_i),$$

where $\alpha \in \mathbb{R}^n$ are the coefficients.

Representer Theorem: Intuition

Goal:

$$\min_{f \in \mathcal{H}} \sum_{i=1}^{n} L(y_i, f(x_i)) + \lambda \|f\|_{\mathcal{H}}^2.$$



Key insight:

- The loss depends on f **only through its values on training data:** $f(x_1), \ldots, f(x_n)$.
- Any part of f orthogonal to $\{\phi(x_1), \ldots, \phi(x_n)\}$ doesn't change these values— it only increases $\|f\|_{\mathcal{H}}^2$.

Therefore, the optimal f must lie in the **span** of $\{\phi(x_i)\}$:

$$f^*(\cdot) = \sum_{i=1}^{n} \alpha_i \, k(x_i, \cdot), \quad k(x_i, \cdot) = \phi(x_i)^\top \phi(\cdot).$$

Kernel Trick (Example Continued)

Feature vectors: $\phi(x_1) = \begin{bmatrix} 1 \\ 2 \\ 1 \\ 2 \\ 4 \end{bmatrix}, \quad \phi(x_2) = \begin{bmatrix} 3 \\ 0 \\ 9 \\ 0 \\ 0 \end{bmatrix}, \quad \phi(x_3) = \begin{bmatrix} -1 \\ 1 \\ 1 \\ -1 \\ 1 \end{bmatrix}$

Kernel: $K(x_i, x_j) = \phi(x_i)^\top \phi(x_j)$
Compute kernel matrix:

$$K = \begin{bmatrix} K_{11} & K_{12} & K_{13} \\ K_{21} & K_{22} & K_{23} \\ K_{31} & K_{32} & K_{33} \end{bmatrix} = \begin{bmatrix} 26 & 12 & 4 \\ 12 & 90 & 6 \\ 4 & 6 & 5 \end{bmatrix},$$

From Vanilla Features to Kernels

- Recall that we have $XX^\top \in \mathbb{R}^{n\times n}$ and $X^\top X \in \mathbb{R}^{d\times d}$ where

$$X = \begin{bmatrix} — & x_1^\top & — \\ — & x_2^\top & — \\ & \vdots & \\ — & x_n^\top & — \end{bmatrix}$$

- With feature maps $\phi : \mathbb{R}^d \to \mathbb{R}^p$ we define a new matrix of transformed features:

$$\Phi = \begin{bmatrix} — & \phi(x_1)^\top & — \\ — & \phi(x_2)^\top & — \\ & \vdots & \\ — & \phi(x_n)^\top & — \end{bmatrix} \quad \text{and } K = \Phi\Phi^\top \in \mathbb{R}^{n\times n},$$

with $K_{ij} = \phi(x_i)^\top \phi(x_j)$.

Covariance or Inner Product

- If we compute

$$\Phi^\top \Phi \in \mathbb{R}^{p \times p}$$

  this is the covariance (or inner product) matrix of features, not the kernel over samples.
- It is $p \times p$ since now we are comparing features to each other.
- This is exactly why in kernel methods we usually work with $K = \Phi\Phi^\top$; i.e. we care about sample similarities not feature similarities.

Kernel Method

- **Kernel matrix:** Symmetric $Q \in \mathbb{R}^{n\times n}$ is positive semi-definite (PSD) with entries:

$$Q_{ij} = \phi(x_i)^\top \phi(x_j), \qquad i, j = 1, \dots, n$$

- **Kernel function:**

$$K(x, z) = \phi(x)^\top \phi(z)$$

  in this notation, entries of kernel matrix are

$$Q_{ij} = K(x_i, x_j)$$

- **Kernel methods:** (sometimes also called "kernel trick")
  - algorithms/methods that use $K(x, z)$ and Q, and avoid using $\phi(x)$ or X directly

Nonlinear Feature Maps

- Suppose we want to do feature mappings

$$f(x) = \theta^\top \phi(x), \quad \phi : \mathbb{R}^d \to \mathbb{R}^p$$

- **Kernel corresponding to ϕ:** we'll see to solve the learning problem/make predictions, we only need to be able to compute

$$K(x, z) = \phi(x)^\top \phi(z)$$

Example: Polynomial Kernel

- Consider $K(x, z) = (x^\top z)^2$
- $\phi : \mathbb{R}^2 \to \mathbb{R}^4$ is

$$\phi(x) = (x_1^2, x_1x_2, x_2x_1, x_2^2)$$

Example: Gaussian Kernel

$$K(x, z) = \exp(-\gamma \|x - z\|^2)$$

aka *radial basis function*. Some observations:

- non-linear kernel with a lot of flexibility
- corresponds to an infinite dimensional ϕ (i.e., cannot implement the corresponding feature mapping ϕ.

Model Fitting by Regularized Least Squares

Consider the ℓ_2-regularized Least Squares problem, aka **Ridge Regression**, or Tikhonov regularization:

$$\min_{\theta} \|X\theta - y\|_2^2 + \lambda\|\theta\|_2^2$$

- y is the n-vector with entries $y_1, \ldots, y_n$

Kernel Ridge Regression: Primal Form

Setup

- Feature map: $\phi(x) \in \mathbb{R}^d$
- Design matrix: $\Phi \in \mathbb{R}^{n \times d}$, i-th row is $\phi(x_i)^\top$
- Labels: $y \in \mathbb{R}^n$

Representer Form: $\theta = \Phi^\top \alpha$

Representer theorem (finite-dimensional version)

This is the representer theorem in action. The optimal parameter vector lies in the span of the feature vectors:

$$\hat{\theta} \in \operatorname{span}\{\phi(x_1), \ldots, \phi(x_n)\}.$$

Kernel Ridge Regression Loss in Terms of α

Solving for $\hat{\alpha}$ and Predictions

Primal vs Dual Views of Kernel Regression

Primal (Feature Space)

- Parameters $\theta \in \mathbb{R}^p$.
- Optimize:

$$\min_{\theta} \|y - \Phi\theta\|^2 + \lambda\|\theta\|^2.$$

- Solution:

$$\hat{\theta} = (\Phi^\top \Phi + \lambda I_p)^{-1}\Phi^\top y.$$

- Expensive if p (feature dim) is large or infinite.

Dual (Data Space)

- Parameters $\alpha \in \mathbb{R}^n$.
- Equivalent optimization:

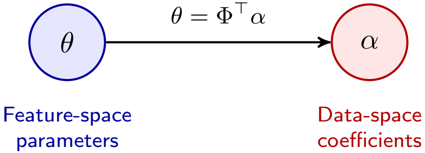
$$\min_{\alpha} \|y - K\alpha\|^2 + \lambda\alpha^\top K\alpha.$$

- Solution:

$$\boxed{\hat{\alpha} = (K + \lambda I)^{-1}y.}$$

- Works via the kernel matrix $K_{ij} = k(x_i, x_j)$—no explicit ϕ needed!

Primal vs Dual Views of Kernel Regression



Connection: feature-space weights θ are expressed via data-space coefficients α.

Why Regularize in Kernel Methods?

Problem without regularization:

- Kernel methods (e.g., kernel ridge regression):

$$f(x) = \sum_{i=1}^{n} \alpha_i K(x_i, x), \quad \alpha = K^{-1}y \quad (\text{if } \lambda = 0)$$

- Issues:
    1. **Overfitting:** Flexible kernels (like RBF) can interpolate training points perfectly but generalize poorly.
    2. **Numerical instability:** K may be nearly singular or ill-conditioned.

Effect of Regularization

Regularized kernel ridge regression:

$$\alpha = (K + \lambda I_n)^{-1} y, \quad f(x) = \sum_{i=1}^{n} \alpha_i K(x_i, x)$$

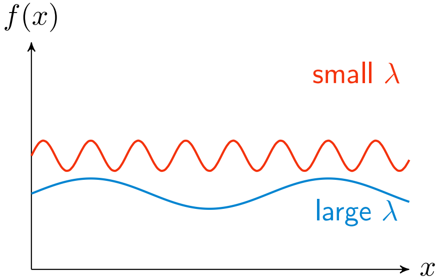
What regularization does:

- Adds λI_n to stabilize matrix inversion.
- Controls the **complexity/smoothness** of the function.
- Small $\lambda \rightarrow$ more flexible (can overfit)
- Large $\lambda \rightarrow$ smoother function, less variance

Regularization in Kernel Methods: Intuition

Key idea: Regularization controls function complexity in feature space.

- Kernel regression: $f(x) = \sum_{i=1}^{n} \alpha_i K(x_i, x)$
- Regularized: $\alpha = (K + \lambda I_n)^{-1} y$
- λ penalizes the RKHS norm $\|f\|_{\mathcal{H}}^2$ (how "wiggly" f is)
- Small $\lambda \rightarrow$ flexible, risk of overfitting
- Large $\lambda \rightarrow$ smoother function, less variance



RKHS perspective:

$$\min_f \sum_{i=1}^{n} (y_i - f(x_i))^2 + \lambda \|f\|_{\mathcal{H}}^2$$

Intuition

- Regularization balances **bias-variance tradeoff** in kernel space.
- Especially important for highly flexible kernels (e.g., RBF or high-degree polynomial kernels).
- Prevents wild oscillations and improves generalization.

Recap: Kernel Method

- Essence of method:
  - If we can write an algorithm only in terms of $\phi(x_i)^\top \phi(x_j)$, we don't need to explicitly enumerate $\phi(x_i)$ for $i = 1, \ldots, n$
  - Instead can just compute $Q_{ij} = K(x_i, x_j) = \phi(x_i)^\top \phi(x_j)$
  - And can even handle infinite-dimensional feature maps $\phi(\cdot)$
- Examples:
  - Linear kernel $K(x, z) = z^\top x$
  - radial basis function $K(x, z) = \exp(-\gamma \|x - z\|^2)$
  - polynomial kernels $K(x, z) = (x^\top z + c)^p$
- Predictions only depend on training data through kernel function (which is just a dot product)

Extra Slides: Representer Theorem

Representer Theorem: Why the Solution Lives in a Span

Goal: Explain why the KRR solution has the form

$$f^*(x) = \sum_{i=1}^{n} \alpha_i k(x_i, x).$$

Key idea: The optimizer lives in the span of the *feature vectors* $\phi(x_1), \ldots, \phi(x_n)$, even though we later express it using kernels.

Setting: Let $f(x) = \langle \theta, \phi(x) \rangle$ in an RKHS $\mathcal{H}$.

$$\min_{f \in \mathcal{H}} \sum_{i=1}^{n} L\left(y_i, f(x_i)\right) \; + \; \lambda \|f\|_{\mathcal{H}}^2$$

The variable is θ (possibly infinite-dimensional).

Step 1: Decompose θ Into Parallel & Orthogonal Parts

Step 2: Regularization Forces $\theta_\perp = 0$

Step 3: Move From Feature Span to Kernel Expansion

Conclusion